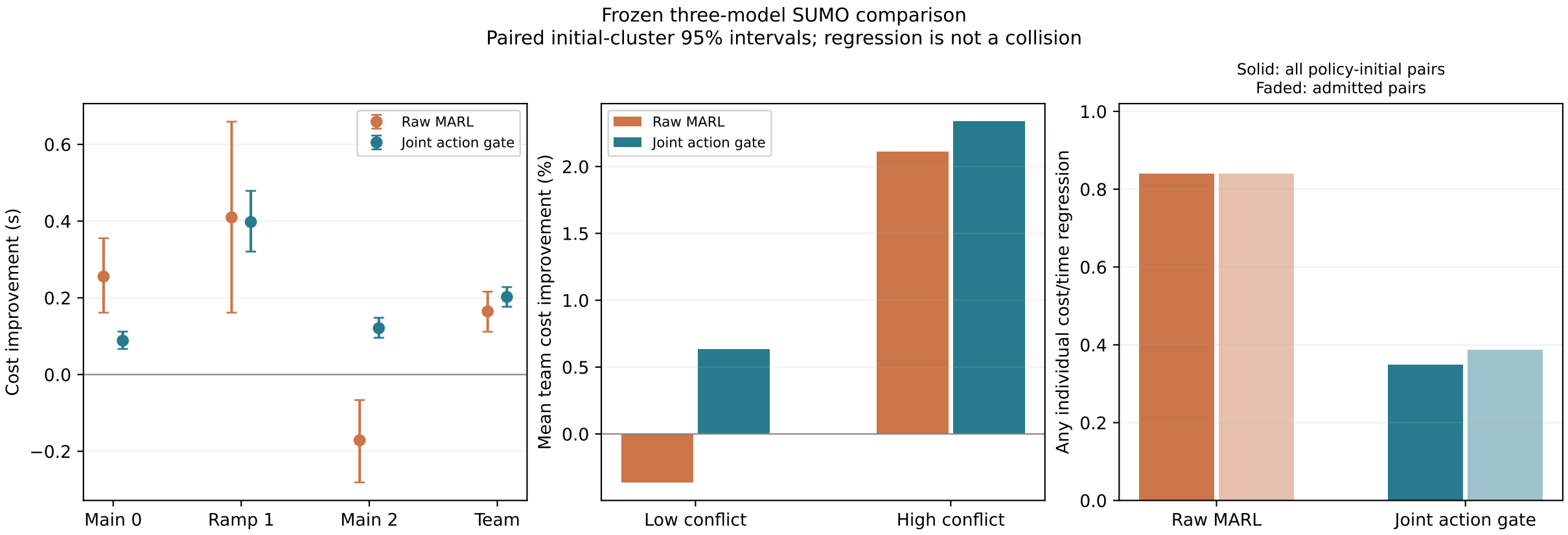
Frozen three-model SUMO comparison
Paired initial-cluster 95% intervals; regression is not a collision
Raw MARL
Joint action gate
Cost improvement (s)
0.6
0.4
0.2
0.0
−0.2
Main 0
Ramp 1
Main 2
Team
Mean team cost improvement (%)
2.0
1.5
1.0
0.5
0.0
Low conflict
High conflict
Solid: all policy-initial pairs
Faded: admitted pairs
Any individual cost/time regression
1.0
0.8
0.6
0.4
0.2
0.0
Raw MARL
Joint action gate